Horizontal and vertical lines

1 State whether the graphs of the following lines are horizontal or vertical:

a $y = 2$ **b** $x = -4$ **c** $y = 3$ **d** $x = 17$

e $y = 0$ **f** $x = 0$ **g** $y - \frac{1}{2} = 0$ **h** $\frac{4}{7} = x$

2 The table shows some points on the line with equation $x = 0$.

Does the line $x = 0$ represent the y-axis or x-axis?

x	0	0	0	0
y	-8	-1	4	6

3 The table shows some points on the line with equation $y = 0$.

Does the line $y = 0$ represent the y-axis or x-axis?

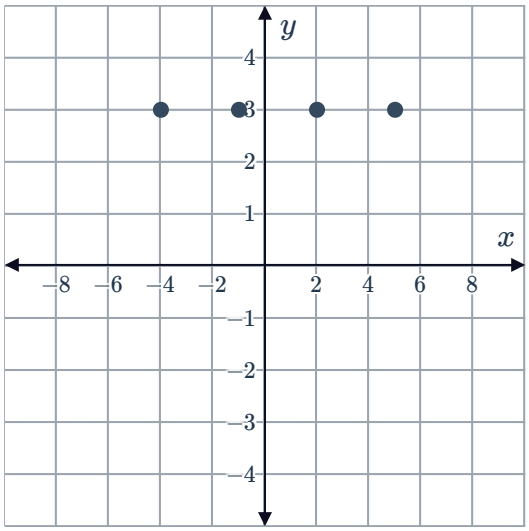
x	-6	-4	1	5
y	0	0	0	0

4 Consider the sets of points in the following coordinate planes:

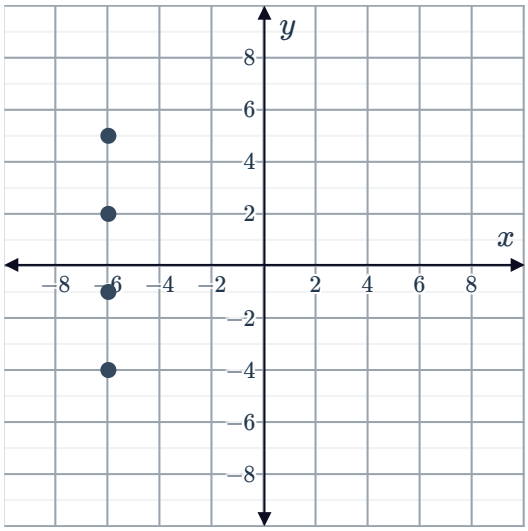
i State whether the set of points lies on a increasing, decreasing, vertical or horizontal line.

ii Find the equation of the line that passes through the set of points.

a

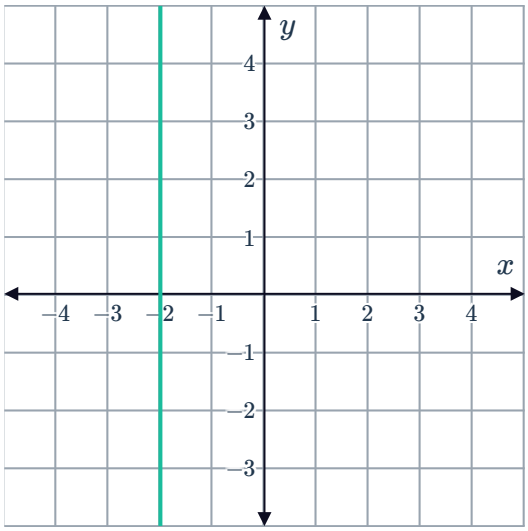


b



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5 Consider the graphed line:

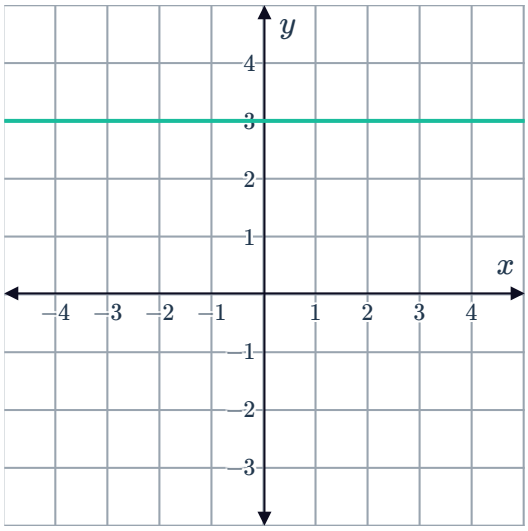


a State the x-intercept.

b Does this line have a y-intercept?

c Write down the equation of the line.

6 Consider the graphed line:



a State the y-intercept.

b Does this line have an x-intercept?

c Write down the equation of the line.

7 For each of the following tables of values:

i Sketch the graph of the line that passes through the four points.

ii Find the equation of the line.

a

x	2	2	2	2
y	-2	-1	0	1

b

x	-4	-2	0	2
y	5	5	5	5

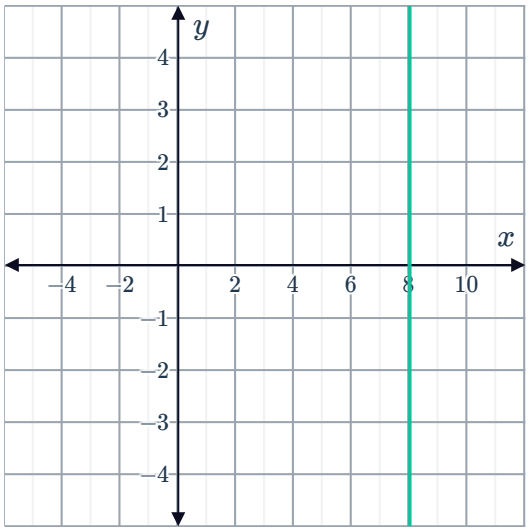
8 Plot the following lines on a coordinate plane:

a $y = 8$ **b** $x = 7$ **c** $y = -2$ **d** $y = 0$

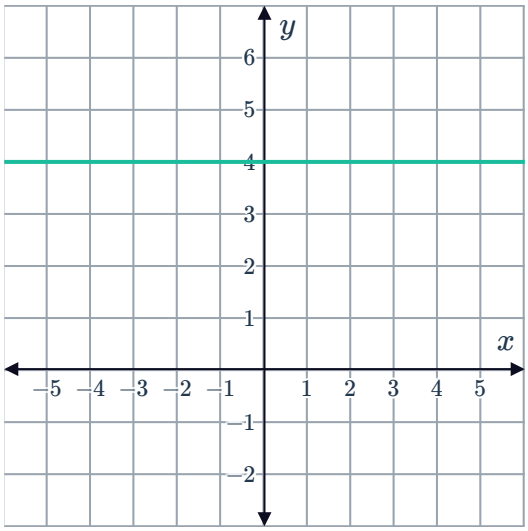
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9 Find the equation of the following lines:

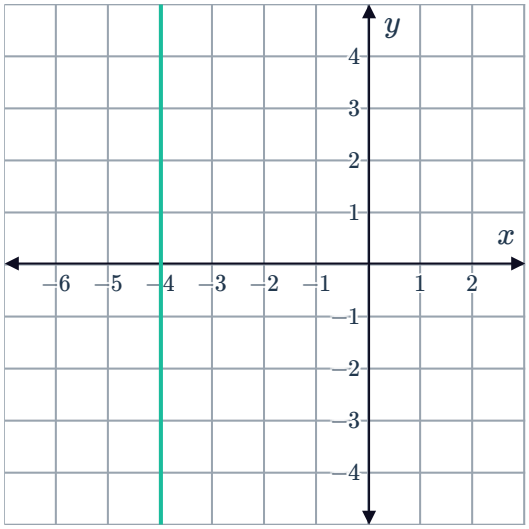
a



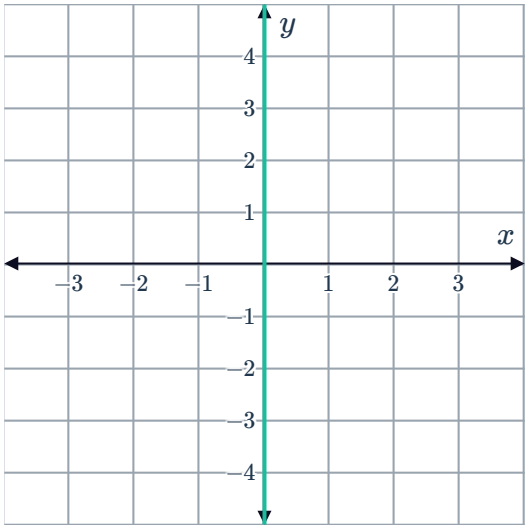
b



c



d



10 Find the equation of the line given the following conditions:

a The line is parallel to the x-axis and passes through the point $(3, -5)$.

b The line is parallel to the y-axis and passes through the point $(-8, 3)$.

c The line is perpendicular to the x-axis and passes through the point $(3, -8)$.

d The line is perpendicular to the y-axis and passes through the point $(3, -4)$.

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11 The temperature (in degrees Celsius) inside a freezer over a five-hour period was recorded:

Time (h)	0	1	2	3	4	5
Temperature (T)	-9	-9	-9	-9	-9	-9

a Sketch the line that passes through each of the pairs of values from the table.

b Write an equation for the temperature T, at time h.

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